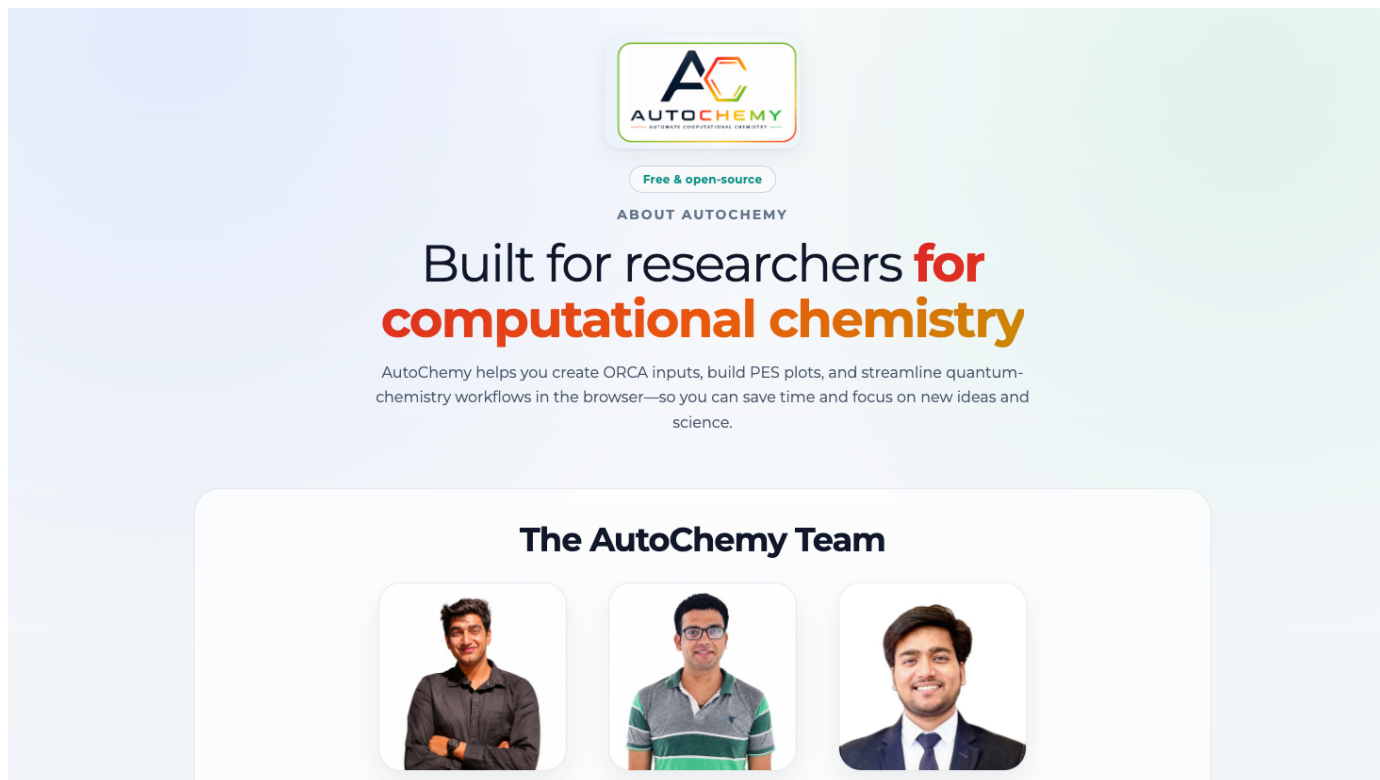


AUTOCHEMISTRY

Quantum-Chemistry Workflows, Made Visual

A concise overview of AutoChemistry's tools for ORCA input creation, calculation setup, xTB screening, reaction-pathway management, potential-energy analysis, and live browser access.

Free and open source | Built for computational chemistry researchers | autochemistry.xyz | [GitHub](#)



The screenshot shows the AutoChemistry website homepage. At the top center is the AutoChemistry logo, which consists of a stylized 'AC' in a hexagonal shape with the text 'AUTOCHEMISTRY' and 'AUTOMATE COMPUTATIONAL CHEMISTRY' below it. Below the logo is a button that says 'Free & open-source'. Underneath that is the text 'ABOUT AUTOCHEMISTRY'. The main heading reads 'Built for researchers for computational chemistry', with 'for' in red and 'computational chemistry' in orange. Below this is a paragraph: 'AutoChemistry helps you create ORCA inputs, build PES plots, and streamline quantum-chemistry workflows in the browser—so you can save time and focus on new ideas and science.' Further down is a section titled 'The AutoChemistry Team' featuring three circular portrait photos of the team members: a man with dark hair in a dark shirt, a man with glasses in a green and grey striped shirt, and a man with dark hair in a suit and tie.

Create inputs. Run and inspect calculations. Build clear mechanistic figures.

FEATURE 01

ORCA Input & HPC Script Generation

Build ORCA calculations from guided controls while AutoChemistry generates both the input file and a reusable shell script beside the molecular structure.

Sam's AutoChemistry

Projects: Project: IITK Sub-project: Bp-86_Opt-Freq Save current... Delete sub Delete project Stores all tabs, geometry, previews, constraints, and file picker paths.

Input & Script Generator Batch Process Data Tools

Theory Method & Basis Parameters Script xTB Pre-submit

Theory Type
DFT

Spin configuration: RKS # closed-shell (RKS for DFT) Auto-Recommend

Job Type
Scan Edit Custom Jobs... Bond Scan

Atoms: 1 4 Pick atoms...
Start (Å/deg): 1 End (Å/deg): 2.5 Steps: 10

System & Output
Charge: 0 Multiplicity: 1
Job Name: water_opt

Geometry (XYZ format)
3
Geometry
O -2.031050 0.000000 -1.318670
H -2.031050 0.759337 -0.722629
H -2.074330 -0.640384 -2.039710

Load XYZ Visualize Clear
Open External Viewer Generate Preview Save Files

Structure visualization
Visualization Software: ACV (AutoChemistryViewer) Open Detach viewer window
Right drag: rotate, wheel: zoom, left click: select
Labels Numbers

Input (.inp)
!RKS BP86 def2-SVP D3BJ RI def2/J NumFreq TightSCF
%scf maxiter 1000 end
%geom maxiter 1000 end
%maxcore 1000
%pal nprocs 3 end
* xyz 0 1
O -2.031050 0.000000 -1.318670
H -2.031050 0.759337 -0.722629
H -2.074330 -0.640384 -2.039710

Job Script (.sh)

Guided calculation setup: Configure theory, spin, job type, scan coordinates, geometry, convergence settings, and advanced ORCA keywords in one workspace.

Local or HPC: Generate ORCA .inp and shell .sh files for a workstation or adapt the script for an HPC scheduler and cluster environment.

FEATURE 02

Functional & Basis-Set Selection

Choose density functionals and basis sets from organized scientific libraries instead of manually recalling and typing every ORCA keyword.

The screenshot displays the AutoChemY software interface. On the left is a sidebar with navigation buttons: Input Creator, Output Viewer, PES Plot, DIA, Orbital Creator, ML, xtb & Conformational Analysis, and About us. The main window is titled 'Input & Script Generator' and contains tabs for Theory, Method & Basis, Parameters, Script, and xTB Pre-submit. The 'Method & Basis' tab is active, showing 'Selected Theoretical Model' as BP86 and 'Basis Details' with 'Basis Set' as def2-SVP. Below this, there's a 'Split Basis Set (By Atom/Element)' section with a target 'Co (0)' and a 'newto' assignment. The 'Approximations' section includes 'Dispersion Correction' and 'Integration Grid'. At the bottom, there's a 'Geometry (XYZ format)' section with a table of coordinates and buttons for 'Load XYZ', 'Visualize', 'Clear', 'Open External Viewer', 'Generate Preview', and 'Save Files'. A 'Select Density Functional' dialog box is open on the right, showing a grid of functionals categorized by GGA, Hybrid & Meta GGA, Hybrid GGA, and Meta GGA. Each functional card displays its name and exact exchange percentage. The dialog also has a 'Grouping' section with radio buttons for Popular / Famous, By Family, and By Developer, and a search bar.

GGA	Hybrid & Meta GGA	Hybrid GGA	Meta GGA
PBE Pure (0%)	M06 Exact Exchange: 27%	B3LYP Exact Exchange: 20%	M06-L Pure (0%)
BLYP Pure (0%)	M06-2X Exact Exchange: 54%	B3LYP/G Exact Exchange: 20%	TPSS Pure (0%)
BP86 Pure (0%)	PW6B95 Exact Exchange: 28%	PBE0 Exact Exchange: 25%	revTPSS Pure (0%)
OLYP Pure (0%)	TPSSH Exact Exchange: 10%	BHLYP Exact Exchange: 50%	
		O3LYP Exact Exchange: 11.6%	
		B3PW Exact Exchange: 20%	

Searchable functional library: Browse popular functionals or group them by family or developer, with useful method information shown directly in the selector.

Flexible basis assignment: Select a main basis set and apply split basis sets to specific atoms or elements for tailored computational protocols.

FEATURE 03

Run ORCA Locally or Use Machine Profiles

Detect a local ORCA installation, configure computational resources, submit calculations directly, or preserve reusable machine settings for scripted HPC workflows.

Sam's AutoChemistry

Projects: Project: Sub-project: Save current... Delete sub Delete project

Input & Script Generator Batch Process Data Tools

Theory Method & Basis Parameters **Script** xTB Pre-submit

Saved Machine: Add new machine... Edit Delete

Local

Local ORCA Runtime Status

6.1.1 | ORCA installed and ready. Version: 6.1.1 (i)

You can run local jobs now.

Cores:

Detected 12 logical cores, ~16254 MB RAM. Recommended local: 3 cores, MaxCore 1000 MB.

Auto Recommend (Local)

Submit Local Job Stop Open Job Folder

Visualize .out Clear Log History

Local job finished.

Timings for individual modules:

Module	Time	Percentage
Sum of individual times	30.525 sec (= 0.509 min)	
Startup calculation	3.310 sec (= 0.055 min)	10.8 %
SCF iterations	22.721 sec (= 0.379 min)	74.4 %
Property calculations	0.593 sec (= 0.010 min)	1.9 %
SCF Gradient evaluation	3.687 sec (= 0.061 min)	12.1 %
Geometry relaxation	0.214 sec (= 0.004 min)	0.7 %

TOTAL RUN TIME: 0 days 0 hours 0 minutes 33 seconds 589 msec

[Done] ORCA exited with code 0

Geometry (XYZ format)

Atom	X	Y	Z
O	-2.031050	0.000000	-1.318670
H	-2.031050	0.759337	-0.722629
H	-2.074330	-0.640384	-2.039710

Load XYZ Visualize Clear

Open External Viewer Generate Preview Save Files

Structure visualization

Visualization Software: Open

Detach viewer window

Right drag: rotate, wheel: zoom, left click: select

Labels Numbers

Run on your computer: AutoChemistry displays the detected ORCA version, available cores, RAM context, and recommended local settings before submission.

Scale to HPC: Saved machine profiles and generated job scripts make repeated cluster submissions more consistent and easier to reproduce.

FEATURE 04

Fast xTB Pre-Screening

Run a quick Lates g-xtb/GFN2-xTB preview before full DFT, inspect the live calculation log, and view a relaxed scan as both numerical energies and a graph.

AutoChemistry

Projects

Project: —Select project— Sub-project: —Select sub-project— Save current... Delete sub Delete project Stores all tabs, geometry, previews, constraints, and file picker paths.

Input & Script Generator Batch Process Data Tools Watch Tutorial

Theory Method & Basis Parameters Script xTB Pre-submit

xTB Quick Preview

⚠ xTB is a semiempirical tight-binding method
Use it for quick screening / geometry pre-check before full DFT.

▶ Run xTB Preview Stop xTB: gfn2 xTB job: Relaxed PES scan (--opt + xcontrol)

Live Log Stream: Open Full Log

xtb_full.log 134,547 bytes
xtbopt.log 138,519 bytes
xtbopt.xyz 5,763 bytes
xtbrestart 6,272 bytes
xtbscan.log 57,410 bytes
xtbtopo.mol 7,280 bytes

Numbers xTB Output Geometry

Scan energies:

Step	Energy (Eh)	dE (kcal/mol)
1	-129.7238938806	34.0772
2	-129.7365314568	26.1470
3	-129.7467222556	19.7522
4	-129.7550319874	14.5377
5	-129.7616244407	10.4009
6	-129.76694131199	7.0631

3D Viewer Scan Graph

Scan graph (kcal/mol)

xTB Relaxed PES Scan - Bond H(71)-H(72)

Relative Energy (kcal/mol)

Bond length H(71)-H(72) [Angstrom]

Use Output XYZ in Main Input Show Scan Graph

XTB output folder: C:\Users\Admin\Desktop\Autochemy-main\Autochemy-main\external_modules\xtb\xtb_runs\xtb_run_8c8kixym
(res.out / xtb_full.log = full stdout; xtbopt.log = opt steps; xtbscan.log = scan; Chemcraft opens trajectory logs when available)

Save xTB files... Clear xTB Open xTB file... Open xTB Folder History Open xTB output external

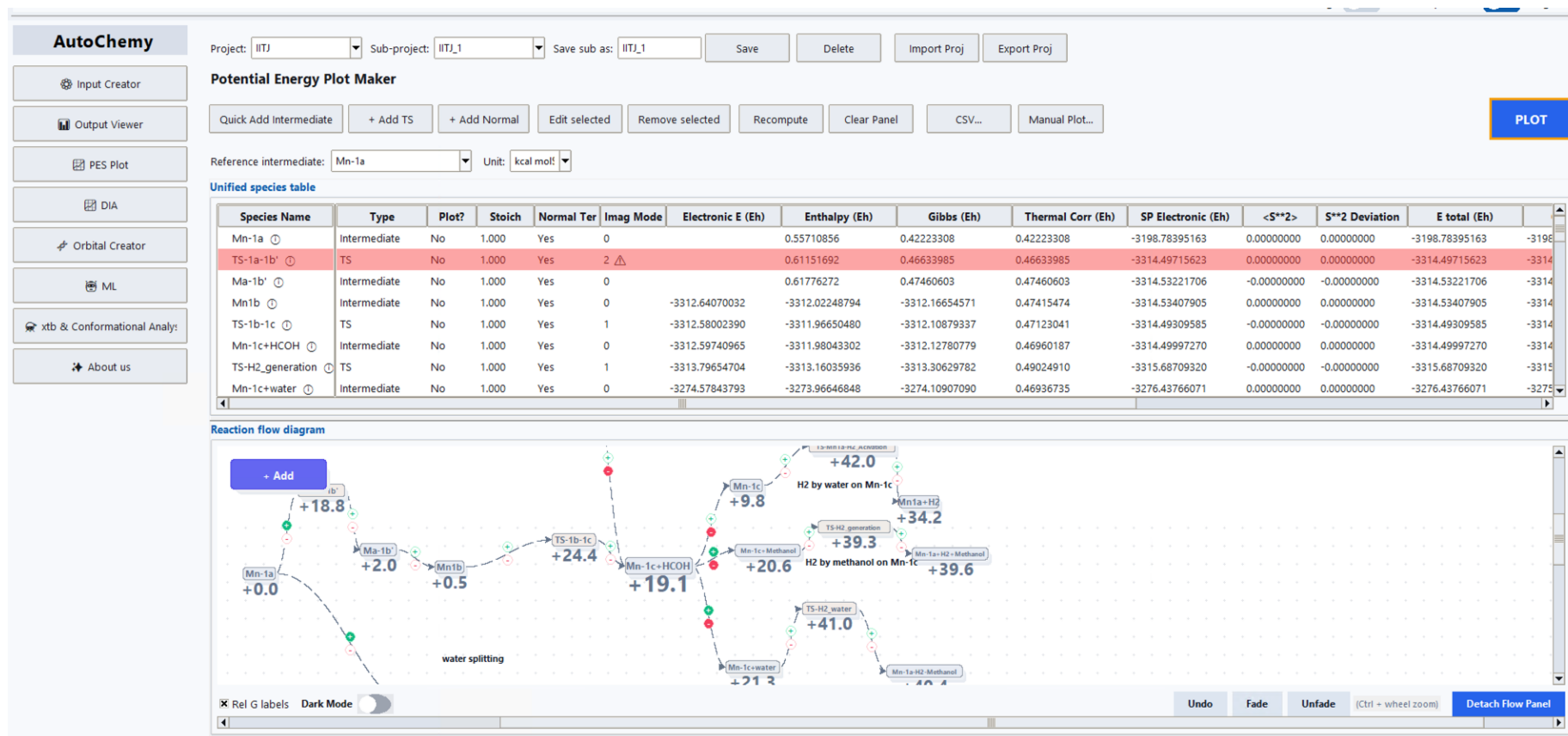
Early insight: Rapid scans help identify useful geometries, trends, and problematic regions before a more expensive calculation.

Direct reuse: The resulting xTB geometry can be transferred back into the main ORCA input workflow.

FEATURE 05

Reaction-Pathway Workspace

Collect intermediates and transition states in a unified energy table, then arrange them as an editable reaction-flow diagram.



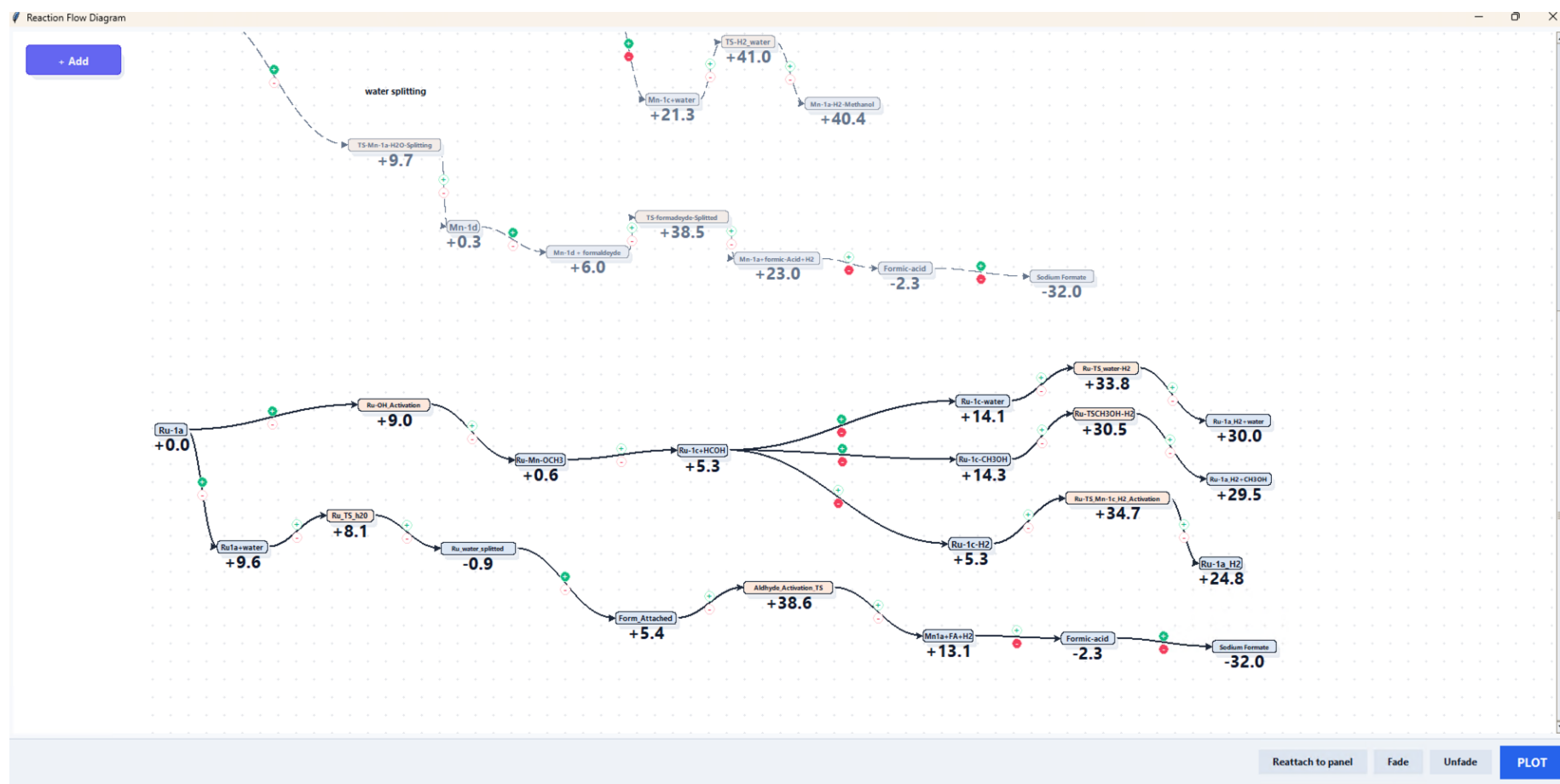
Data in context: Electronic, thermal, enthalpy, Gibbs, and spin-related values remain connected to each species.

Mechanistic organization: Branches and competing pathways can be assembled visually before producing the final PES figure.

FEATURE 06

Detachable Reaction-Flow Canvas

Open the pathway diagram in a large independent window to inspect complex catalytic networks and compare multiple mechanistic branches.



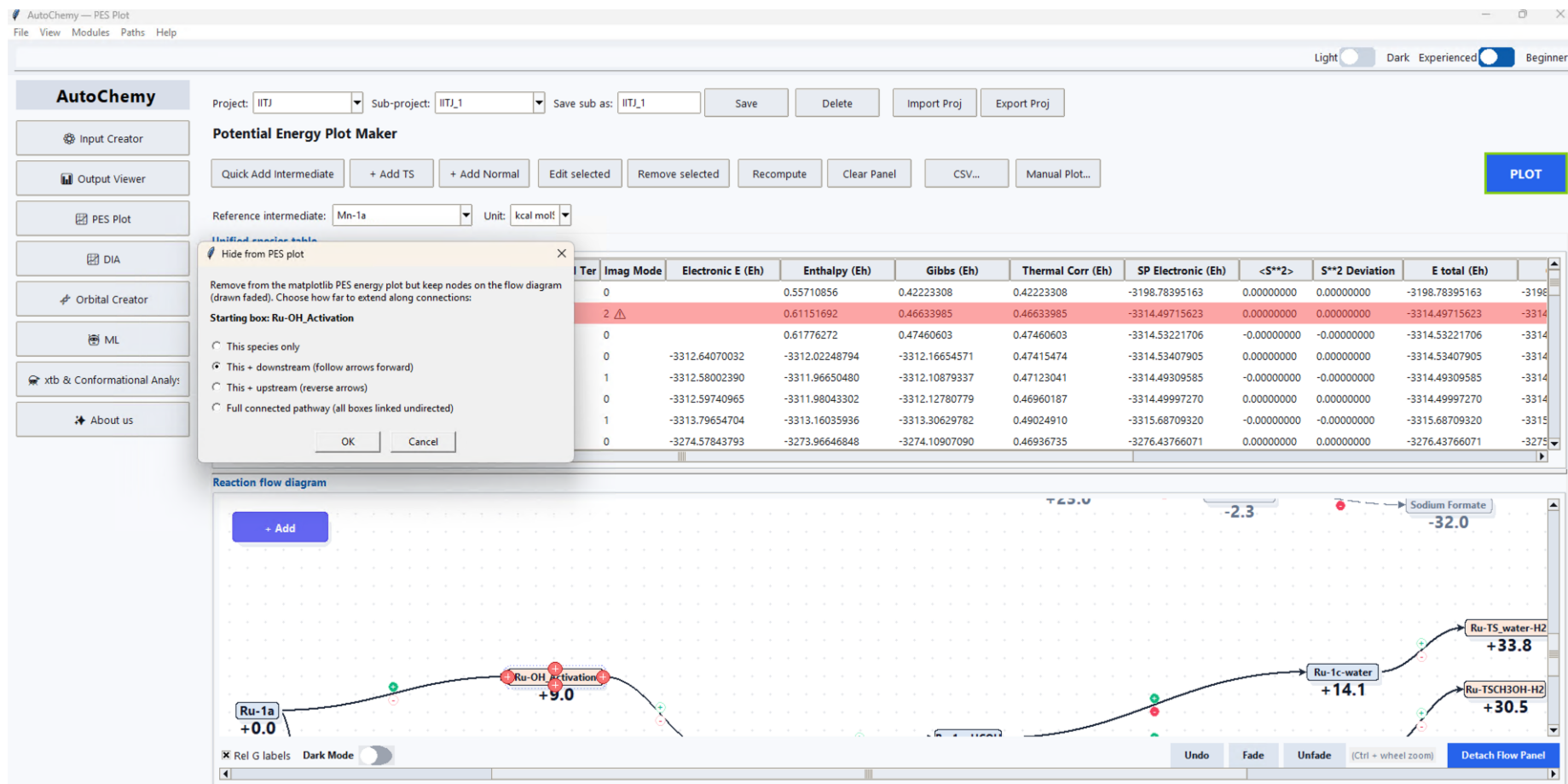
Large-network visibility: A spacious canvas keeps long and branched pathways readable during analysis.

Interactive editing: Species and connections can be extended, faded, restored, and reorganized as the mechanism develops.

FEATURE 07

Selective Pathway Control

Choose exactly which species or connected pathway should appear in the final PES plot while retaining the complete reaction network in the workspace.



Focused figures: Hide one species, a downstream branch, an upstream branch, or an entire connected pathway.

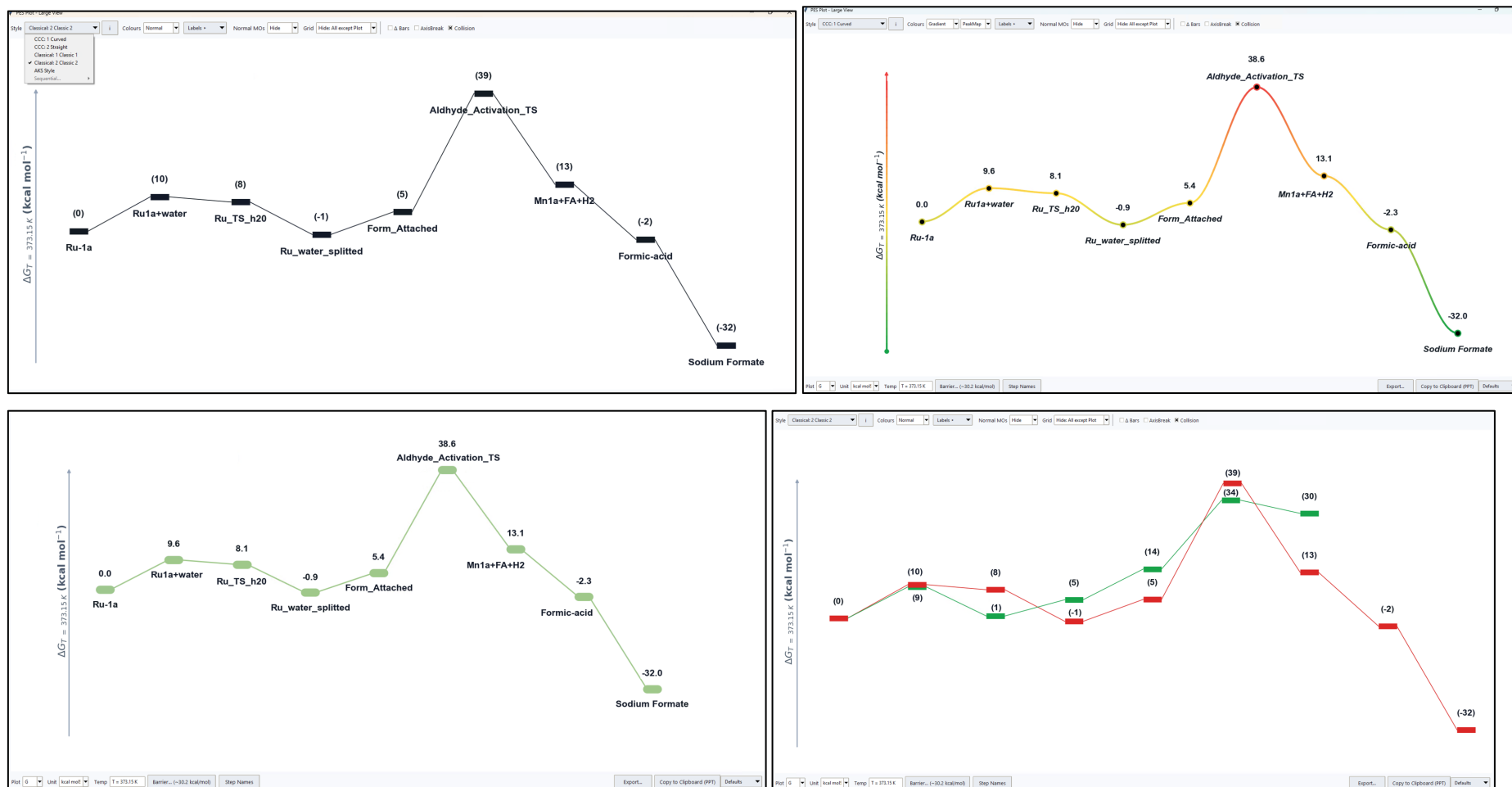
Non-destructive analysis: Nodes remain available in the reaction-flow diagram, so alternate mechanisms are not lost.

FEATURE 08

Publication-Ready Potential-Energy Plots

Generate a clean potential-energy profile from the selected pathway and refine the figure using built-in styles, labels, colors, grids, and axis controls.

Publication-quality plots for easy comparison of relative Gibbs free energies, enthalpies, and electronic energies of intermediates and transition states.



Scientific communication: Consistent energies and species labels turn calculation data into a figure that is ready for discussion or further polishing.

Rapid iteration: Style and visibility controls make it easy to compare figure treatments without rebuilding the plot manually.

FEATURE 09

CREST Conformational Sampling

Prepare and run CREST sampling, inspect logs and ensemble structures, deduplicate conformers, and create downstream ORCA inputs from selected geometries.

The screenshot displays the AutoChemistry xtb & Conformational Analysis interface. The left sidebar contains navigation buttons: Input Creator, Output Viewer, PES Plot, DIA, Orbital Creator, ML, xtb & Conformational Analysis (selected), and About us. The main workspace is titled 'xTB Workspace' and features a 'Conformational Sampling' section. Under 'CREST Conformational Sampling', the 'Geometry (XYZ - atom lines or full XYZ)' section includes a 'No structure loaded' message, a 'Load structure...' button, and input fields for 'Charge: 0' and 'Mult: 1'. Below these are 'Load XYZ...' and 'Save XYZ...' buttons. To the right are 'Run CREST', 'Stop', 'Open result: Chemcraft', and 'Open XYZ file...' buttons. A 'Live Log Stream' window is visible. The 'Post-processing' section includes a 'Deduplicate pool → cret_ensemble.xyz' dropdown set to 'cregen', a description of the 'cregen' command, and checkboxes for 'Split ensemble into individual XYZ files' (checked), 'Generate ORCA inputs per conformer' (unchecked), and 'Create CSV of conformer energies' (checked). An 'Energy window (kcal/mol): 6.0' field is also present. The 'ORCA input settings' section includes fields for 'Method: B3LYP', 'Basis: def2-SVP', and 'Job: Opt', along with 'Extra keywords:' and buttons for 'Import template from input Creator...', 'Open Input Creator', and 'Clear template'. A 'Summary' section is at the bottom left. On the right, the 'Output Geometry / Ensemble Preview' section is empty, and the 'CREST relative energies' section contains the text: 'After CREST: sorted energies, E_{min} (E_h), Boltzmann % @ 298 K, and rank vs CREST file index.'

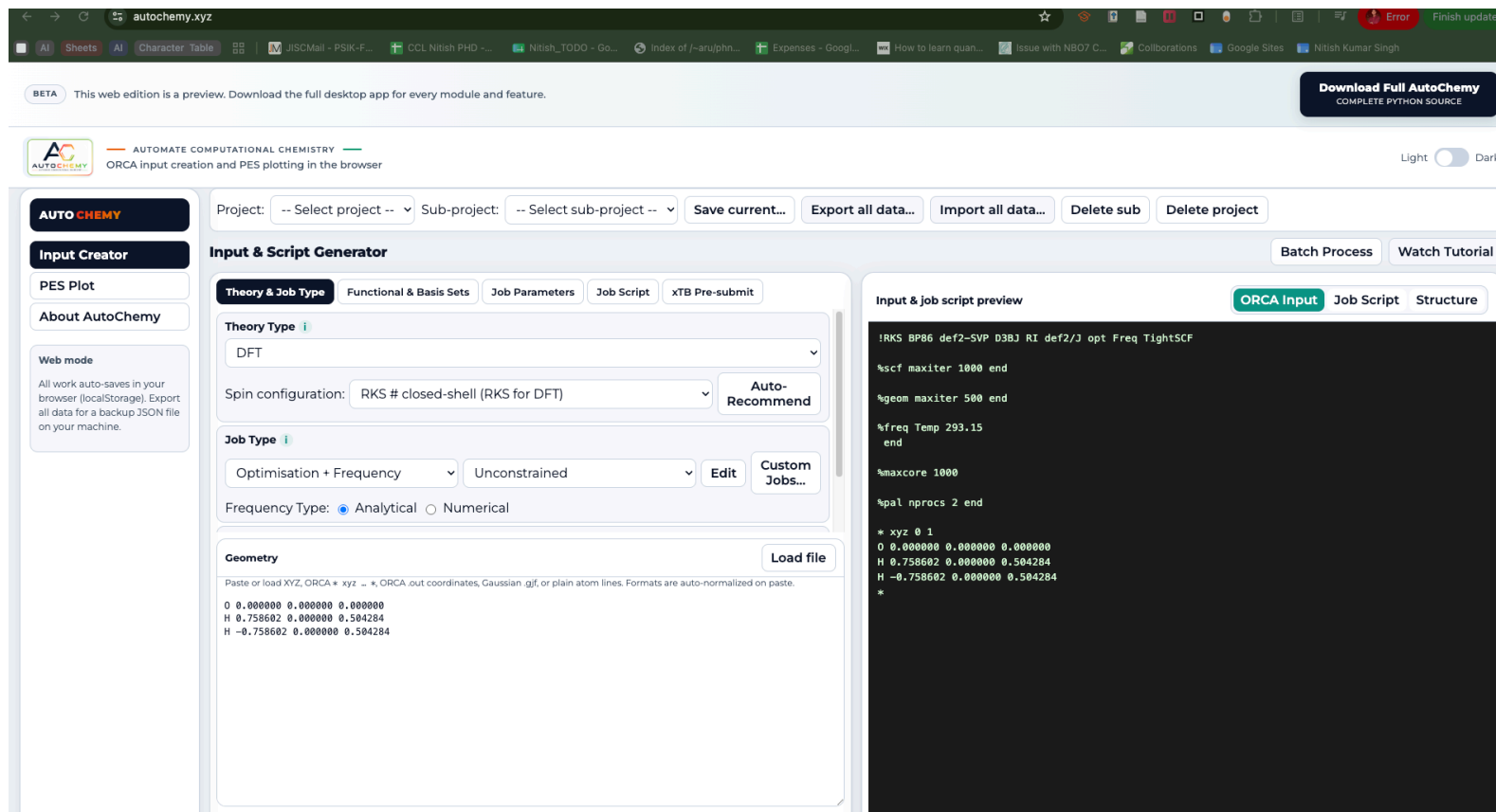
End-to-end sampling: Structure loading, charge and multiplicity, CREST execution, and post-processing are brought together.

Reusable outputs: Export individual XYZ files, conformer-energy summaries, and ORCA-ready inputs for follow-up calculations.

FEATURE 10

Live AutoChemY Web Beta

Open the AutoChemY beta directly at autochemistry.xyz to create ORCA inputs and prepare PES workflows in a browser, with no desktop installation required for the preview modules.



Easy access: The beta is live in the browser, making it convenient to explore AutoChemY, prepare inputs, and demonstrate the interface from any suitable computer.

Browser-safe workflow: Projects auto-save locally in the browser and can be exported or imported as backup data; the complete desktop source remains available for all modules and features.

OPEN THE LIVE BETA autochemistry.xyz

FEATURE 11

Open-Source and Researcher-Led

AutoChemistry is developed by researchers to reduce repetitive quantum-chemistry work and help users spend more time on scientific interpretation.

Motivated by Science

Built by young researchers to support the scientific community, allowing you to save time and focus on what matters most: new ideas and science.

Never Settle for Less

We are committed to ensuring you never have to compromise on quality or features.

Feedback & Support

We are researchers and learning developers doing our best to improve AutoChemistry. If you find bugs, scientific inaccuracies, or unexpected results, please report them so we can fix them for everyone.

Report Bug / Request Feature

Share

AutoChemistry is free and open-source. We would be thrilled to see it used in workshops to teach computational chemistry, or to help researchers save valuable time. Please share it with others so they can benefit from it as well!

Community use: The software is free to share for research, teaching, workshops, and computational-chemistry training.

Feedback loop: Users can report issues or request features, supporting continuous improvement of the scientific workflow.

EXPLORE AND DOWNLOAD autochemistry.xyz | [Full source code on GitHub](#)

FOR MORE DETAILS
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